A CLOSED-SET LLM ATTRIBUTION EXPERIMENT

Who wrote it?

Identifying the source model family from its generated response

TextCNN vs. BiLSTM · matched prompts · measured results

GPT-4-0613 **Claude-2** **Gemini-Pro**

Team leader: Kiseong Kim | Team member: Jisung Park

Four questions: identification, prompt context, transfer, and stylistic signals.

Matched questions make the control meaningful

448
shared prompts

1,344
model responses

429
template components

One question → three answers. Each family has exactly 448 rows.

Training	Validation	Test
278 prompts / 834 rows	83 prompts / 249 rows	87 prompts / 261 rows

Near-duplicate templates stay together. Vocabulary is learned on training data only.
Balanced prompt-only inputs must yield 33.3% accuracy; explicit model self-identities are redacted.

Source: official AlpacaEval outputs; exclusions, licenses, model IDs and checksums are documented.

Two small classifiers, one evaluation protocol

TextCNN

Token IDs → 64-dim embeddings

↓

Kernel sizes 3 / 4 / 5; 48 filters each

↓

Padding-aware max pooling

↓

Dropout 0.3 → 3-family softmax

Bidirectional LSTM

Token IDs → 64-dim embeddings

↓

48 hidden units in each direction

↓

Packed sequences → final hidden states

↓

Dropout 0.3 → 3-family softmax

Fair answer budget: response ≤256 tokens; prompt ≤64; combined ≤321 with separator.
Seeds 42 / 43 / 44; AdamW lr 0.002; batch 32; ≤20 epochs; stop after 4 stale epochs.

Select by validation Macro-F1, then validation loss. Evaluate the restored checkpoint once on the test set.

REQUIRED SIX CONDITIONS

RQ1: response text identifies source families above chance

Architecture / input	Accuracy	Macro-F1
TextCNN · Prompt only	33.3% ± 0.0 pp	29.6% ± 1.2 pp
TextCNN · Response only	73.8% ± 2.5 pp	73.5% ± 2.4 pp
TextCNN · Prompt + response	76.4% ± 1.5 pp	76.0% ± 1.7 pp
BiLSTM · Prompt only	33.3% ± 0.0 pp	32.2% ± 0.6 pp
BiLSTM · Response only	75.4% ± 1.5 pp	75.4% ± 1.5 pp
BiLSTM · Prompt + response	54.2% ± 2.5 pp	54.1% ± 2.4 pp

Mean ± sample SD over three training seeds. Response-only TF-IDF baseline: 79.7%.
All prompt-only neural controls equal the theoretical 33.3% chance accuracy.

PAIRED TEST-CLUSTER COMPARISONS

RQ2: does prompt context help?

TextCNN

+2.6 pp

Combined − response-only accuracy

Paired 95% CI: [-0.5, +5.7] pp

BiLSTM

-21.2 pp

Combined − response-only accuracy

Paired 95% CI: [-26.2, -16.0] pp

CNN's interval includes zero; its observed gain is inconclusive.
Same questions, same seeds, same response budget for both architectures.

Hypothesis: prepended prompts alter final reverse-LSTM states and optimization over longer sequences.
This is not a causal proof; attention or separate encoders would test it.

2,000 paired bootstrap resamples of complete template components; test-sample uncertainty only, with no multiple-comparison claim.

RQ3: evaluate both source shift and writing tasks

Response-only model	Main test	SelfInstruct held out	Writing held out
TextCNN	73.8%	59.9%	72.1%
BiLSTM	75.4%	68.9%	69.6%
TF-IDF + LR	79.7%	65.6%	72.1%

SelfInstruct: verified upstream source; 242 unseen prompts.
Writing: deterministic input-only category; 68 unseen prompts.

Holdouts are exploratory seed42 runs. Main neural results average three seeds.
Training size and test distributions change; differences do not isolate a causal domain effect.

Template components cannot cross train/validation/test; non-target component members are purged when necessary.

RQ4: style and vocabulary carry measurable signals

Style features only

59.4%

Length, diversity, punctuation and formatting (16 features)

Seed42 output	TextCNN	BiLSTM
Raw / 256 tokens	76.2%	74.3%
First 64 tokens	75.9%	73.9%
Formatting normalized	66.3%	68.2%

Training-only lexical associations from TF-IDF

Claude

here / here is / here are / are

GPT

remember / sure / sure here / might

Gemini

such / such as / it is / including

Each condition applies one transformation and retrains; multiple signals can change.
Lexical terms explain the baseline's associations, not CNN/LSTM decisions or universal provider traits.

What we learned — and what we can claim

1. Response text predicts these three model families above chance.

2. Prompt context changes CNN by +2.6 pp and BiLSTM by -21.2 pp.

3. Transfer and style controls reveal useful signals alongside dataset confounds.

Limits: historical model versions; one benchmark; unequal source decoding caps; small matched dataset; no unknown-family or new-version detection claim.

Reproducible package: source + licensed data provenance + 38 measured runs + predictions + checkpoints + report + verification. AI assistance is disclosed; students must understand the work.

Source review: Koala excluded from all splits; all 38 runs retrained. See report and data/NOTICE.md.